

LARGE LANGUAGE MODELS

AI-powered language understanding and generation

TABLE OF CONTENTS

- What are LLMs? 01
- Popular LLMs & Use Cases 02
- Advantages vs Challenges 03
- Future of LLMs 04

WHAT ARE LLMS?

A Large Language Model (LLM) is an advanced predictive foundation model trained on massive text collections via deep learning to understand, generate, and process human-like language.



Large Scale

Utilizes massive training datasets and billions of parameters, enabling robust language capabilities and complex reasoning.



Foundation Models

Serve as adaptable base systems that can be fine-tuned for a wide range of specific downstream tasks and applications.



Multimodal Capabilities

Modern models can process and generate multiple data types beyond text, including images, audio, and code.

MODELS & USE CASES

👑 Leading Foundation Models

The landscape is dominated by powerful systems like **GPT-4o** (OpenAI), **Gemini 1.5** (Google), and **Claude 3** (Anthropic). These models excel at advanced reasoning, long-context processing (up to 1M tokens), and multimodal understanding.

🔗 Open Source & Specialized

Accessible innovation is driven by **LLaMA 3** (Meta) and **Mistral 7B**, which offer efficient open-weight alternatives. Specialized models like **BERT** and **RoBERTa** remain essential for specific NLP embeddings and tasks.

✍️ Content & Creation

LLMs are revolutionizing creative workflows by generating human-like text, creating complex images from descriptions, and translating languages with high accuracy. They enable rapid content scalability across various media formats.

🤖 Technical & Interactive

Beyond creativity, LLMs power technical solutions including writing and debugging code, summarizing massive datasets, and driving sophisticated **chatbots** that handle complex question-answering scenarios.

LATEST POPULAR LLMs IN 2024/2025

A comprehensive overview of the leading Large Language Models available in the market

OPENAI

- GPT-4o
- GPT-4o mini
- GPT-4.5 (Preview)
- o1
- o1-mini
- o3 (Preview)
- o3-mini (Preview)

ANTHROPIC

- Claude 3.5 Opus
- Claude 3.5 Sonnet
- Claude 3.5 Haiku
- Claude 3 Opus
- Claude 3 Sonnet
- Claude 3 Haiku

GOOGLE

- Gemini 1.5 Pro
- Gemini 1.5 Flash
- Gemini 1.5 Pro 002
- Gemini 1.5 Flash 002
- Gemini 2.0 Flash
- Gemini 2.0 Pro (Exp)

META

- Llama 3.1 405B
- Llama 3.1 70B
- Llama 3.1 8B
- Llama 3.1 405B Instruct
- Llama 3.1 70B Instruct
- Llama 3.1 8B Instruct

MISTRAL AI

- Mixtral 8x22B
- Mixtral 8x7B
- Mixtral 8x6.7B
- Mixtral 8x22B Instruct
- Mixtral 8x7B Instruct
- Mistral Large 2 (Preview)
- Mistral Small 3

xAI

- Grok-1.5
- Grok-1.5 6B
- Grok-1
- Grok-2 (Beta)
- Grok-2 mini (Beta)

COHERE

- Command R+
- Command R
- Command R Lite
- Command R 7B
- Command R 7B (Instruct)
- Aya Expansive 32B

PERPLEXITY

- PPLx-70B Online
- PPLx-70B
- PPLx-7B Online
- PPLx-7B

DEEPSEEK

- DeepSeek-V3
- DeepSeek-V2.5
- DeepSeek-R1
- DeepSeek Coder V2
- DeepSeek Math 7B

QWEN (ALIBABA)

- Qwen2.5-Max
- Qwen2.5-72B-Instruct
- Qwen2.5-32B-Instruct
- Qwen2.5-14B-Instruct
- Qwen2.5-7B-Instruct
- Qwen2.5-1.5B-Instruct

BAIDU

- ERNIE 4.0 Turbo
- ERNIE 4.0
- ERNIE 3.5
- ERNIE Speed
- ERNIE Lite
- ERNIE Tiny

MOONSHOT AI

- Kimi k1.5 (200B)
- Kimi k1.5 (Long Context)
- Kimi k1.5 (Preview)
- Moonshot v1 128K
- Moonshot v1 32K

MICROSOFT

Phi-3.5-MoE (42B) Phi-3-medium (14B) Phi-3-mini (3.8B) Phi-3-mini-128K (3.8B) Phi-3-small (7B) Phi-2 (2.7B)

Amazon

Nova Pro Nova Lite Nova Micro Nova Canvas Nova Reel

AI21 LABS

Jurassic-2 Ultra Jurassic-2 Mid Jurassic-2 Mini Jurassic-2 Lite

INFLECTION AI

Inflection-2.5 Inflection-2

UPSTAGE

Solar Pro 2 Solar Mini 2 Solar Pro Solar Mini

OTHER POPULAR OPEN SOURCE LLMs

Falcon 180B Falcon 40B Nous Hermes 3 (70B, 8B) Yi-1.5 (34B, 6B) StarCoder 2 (15B, 7B) Gemma 2 (27B, 9B, 2B) DBRX instruct OpenChat 3.5 (7B) GLM-4 (9B) SmolLM2 (1.7B, 135M) LLaVA-NeXT (7B, 13B, 34B) Qwen2 (72B, 57B, 7B, 1.5B)

...and many more on [Hugging Face](#)

SPECIALIZED / DOMAIN-SPECIFIC LLMs

Code Llama 3 3 (70B, 8B) GPT-4o (Vision) Med-PaLM 2 Bloomberg GPT Claude 3 (Vision) Llama 3.1 (Vision) PaLM 2 Flan-T5 (XXL) Stable Diffusion 3 (LLM) Reka Core & Flash MPT-30B Instruct Dolly 12B

ADVANTAGES & CHALLENGES

01

KEY ADVANTAGES

- **Flexibility:** Performs multiple tasks (translation, summarization) with minimal modification.
- **Fluency:** Generates natural, coherent text remarkably similar to human writing.
- **Scalability:** The same architecture supports both small prototypes and large-scale applications.
- **Integration:** Can be easily deployed through APIs and connected to existing workflows.

02

COMMON CHALLENGES

- **Time-Intensive Training:** Developing large-scale foundational models can take weeks or months.
- **Bias Risks:** Models can inadvertently reflect and amplify biases present in their training data.
- **High Cost:** Training and deploying these massive models require significant computational resources.
- **Security:** Potential for misuse or data leakage requires implementation of robust safeguards.

FUTURE OF LLMS

The landscape of Artificial Intelligence is rapidly evolving. We are moving beyond simple text generation toward integrated, autonomous systems that will fundamentally reshape business operations and workforce dynamics.

-  **Multimodal Integration:** Combining text, image, and audio for holistic understanding.
-  **Agentic AI Systems:** Autonomous reasoning and action-taking capabilities.
-  **Domain-Specific Models:** Smaller, efficient models optimized for privacy and speed.
-  **Workforce Evolution:** Emergence of new roles like prompt engineering and model evaluation.



Use cases

Generating text

Translating text

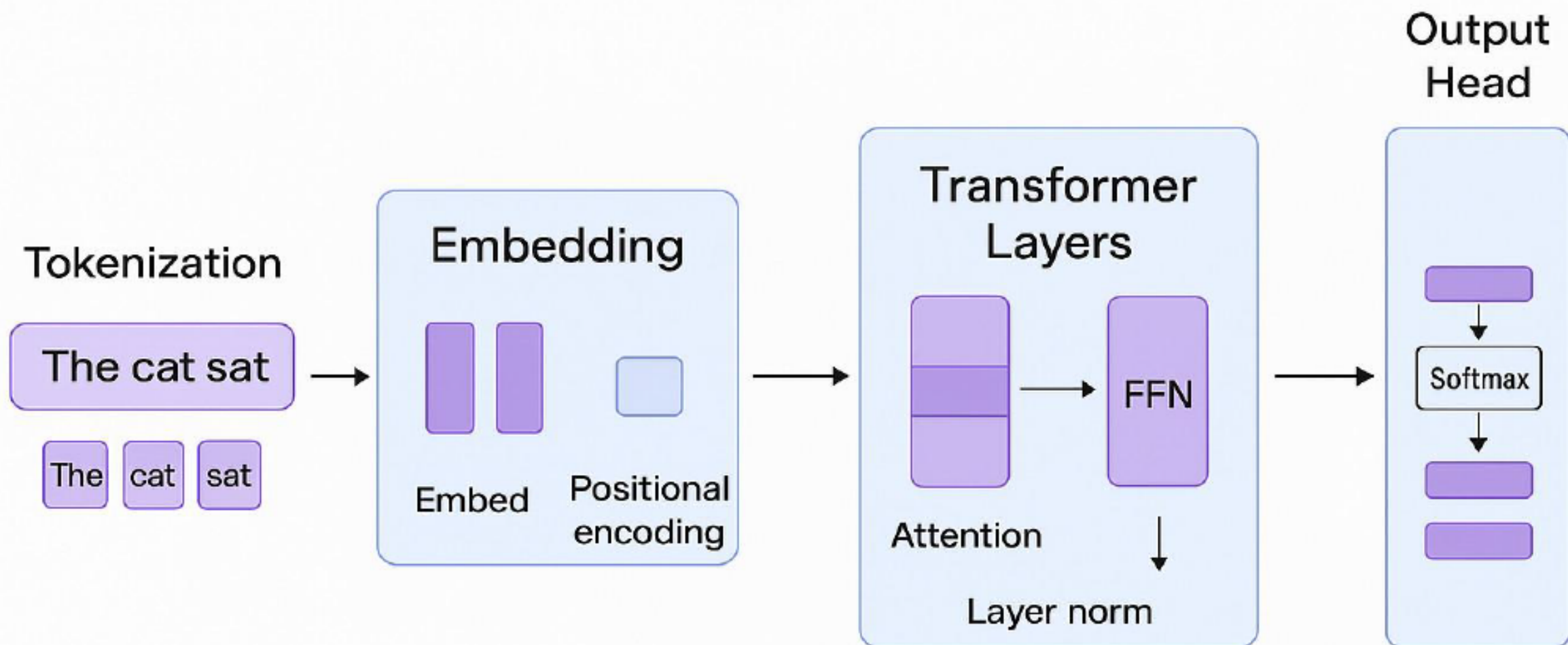
Summarizing data

Writing code

Creating images

Chatbots - question answering.

How Do Large Language Models Work?



Token

- Text input is divided into smaller units called tokens, which can be words or sub words.

(Encode text into numeric rep.)

Dolly is a language model that can help you generate text for a variety of tasks.



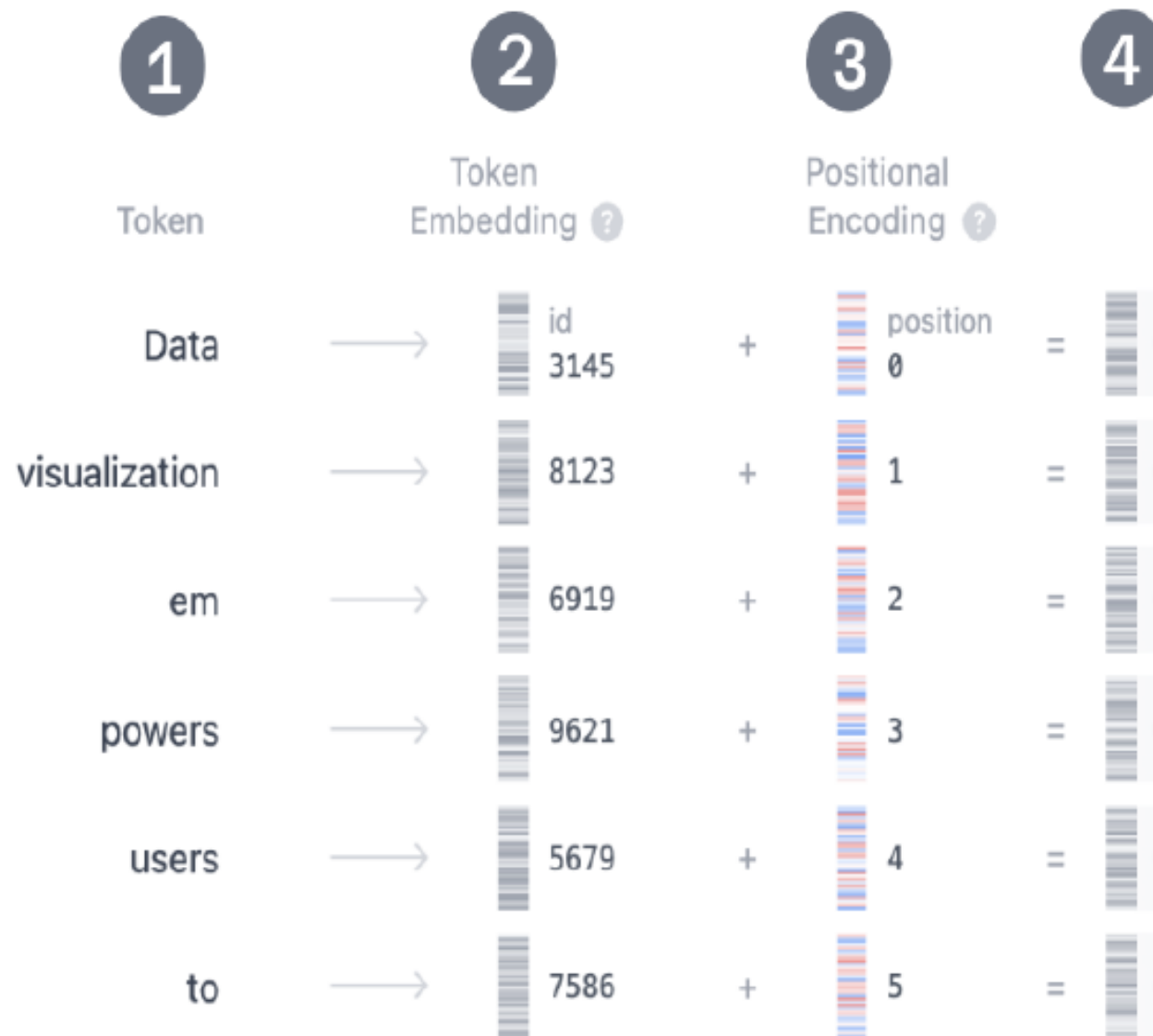
Dolly is a language model that can help you generate text for a variety of tasks.



[35, 5098, 318, 257, 3303, 2746, 326, 460, 1037, 345, 7716, 2420, 329, 257, 4996, 286, 8861, 13]

Prompt:

Data visualization empowers users to ...



Embedding

- “Turns text into Numbers”
- Before a transformer can use the text, it first breaks it into small units and represents each as a list of numbers (Vector) this process is called embeddings, And the term can refer to both the process and resulting vector.

(Put words with similar meaning close in vector space)

Embedding Functions
(Pretrained model)



[0.2, 1.5, 0.6 ... 0.6]

When done well, similar words will be closer in these embeddings/ vector spaces.

Example 2D representation;



Transformers

- **Transformer Block** is the fundamental building block of the model that processes and transforms the input data.
- Each block includes:
 - **Attention Mechanism**, the core component of the Transformer block. It allows tokens to communicate with other tokens, capturing contextual information and relationships between words.
 - **MLP (Multilayer Perceptron) Layer**, a feed-forward network that operates on each token independently. While the goal of the attention layer is to route information between tokens, the goal of the MLP is to refine each token's representation.